

TITLE PAGE

SecureMailScope

AI-Assisted Cryptographic Security Posture Assessment for Secure Email Communications

Problem Statement ID – SIH26159

Problem Statement Title – SecureMailScope: AI-Assisted Cryptographic Security
Posture Assessment for Secure Email Communications

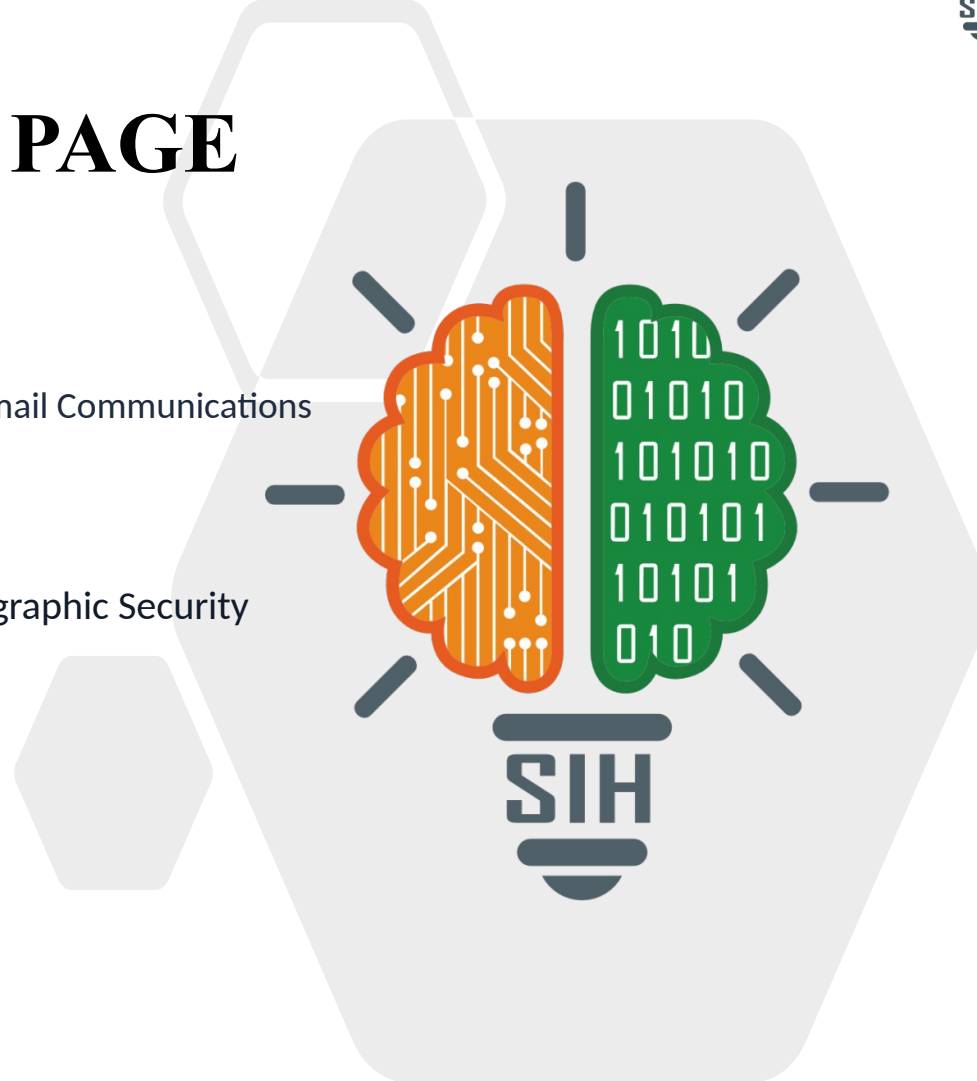
Organisation – National Technical Research Organisation (NTRO)

Theme – Blockchain & Cybersecurity

PS Category – Software

Team ID – 146876

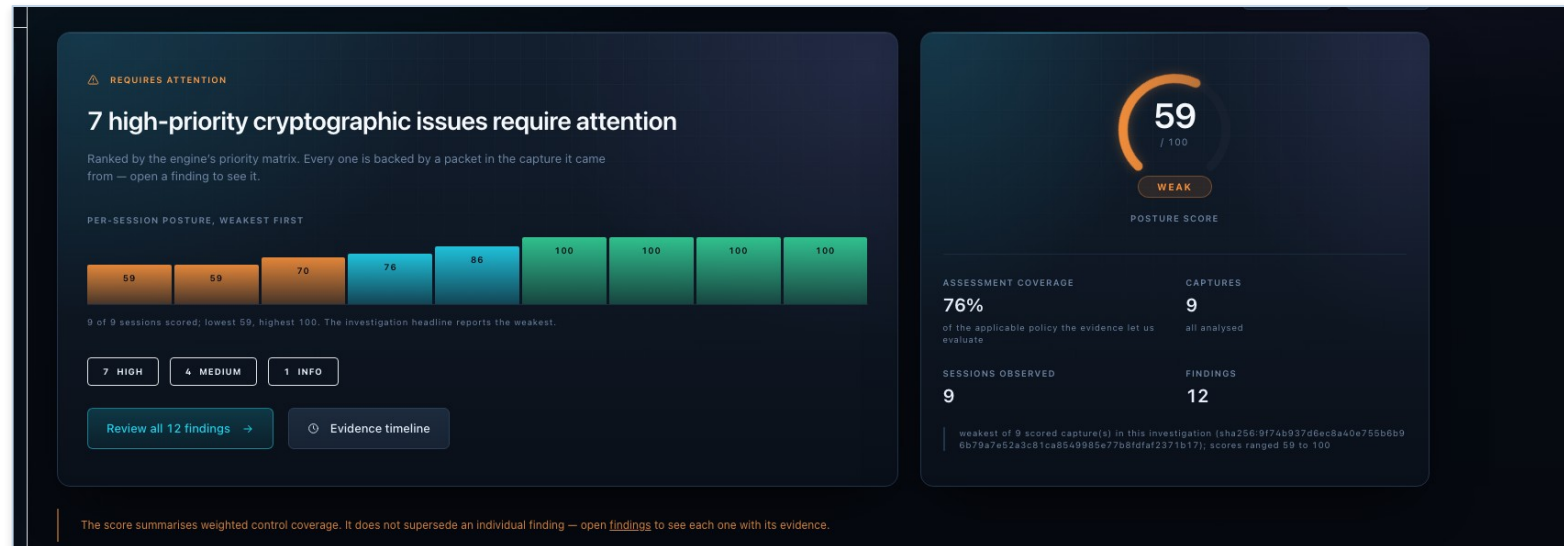
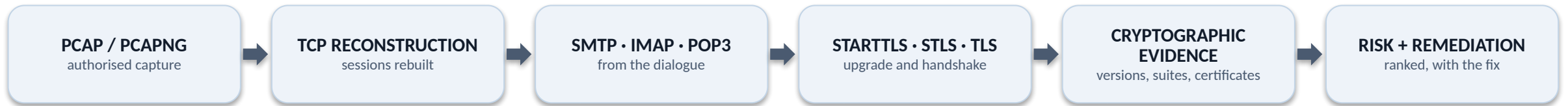
Team Name (Registered on portal) – Zero-Day



Zero-Day

IDEA TITLE

SecureMailScope turns passive email packet captures into evidence-backed cryptographic investigations.



The investigation opens on its conclusion. Actual product output.

PASSIVE BY DESIGN

Existing captures only. No live probe, no connection back to the mail server, nothing leaves the machine.

EVIDENCE-FIRST

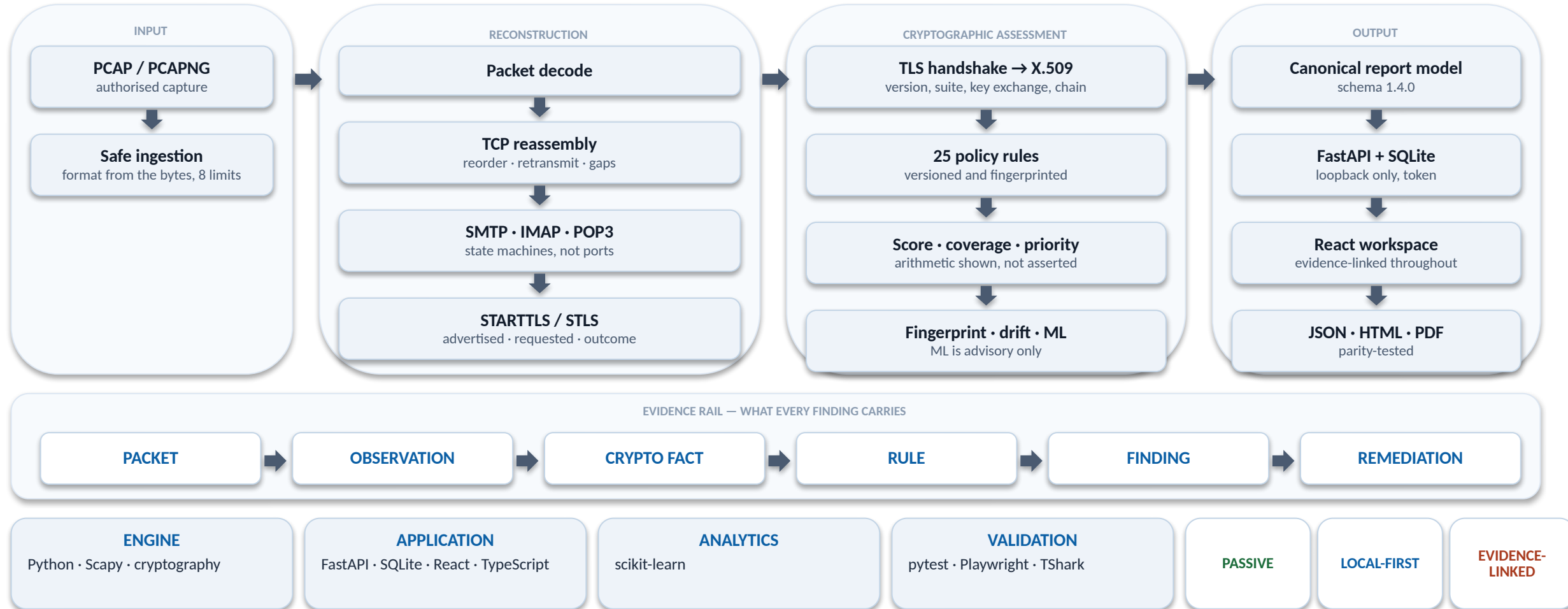
Every finding traces back to the packets that establish it — number, timestamp and stream offset.

CROSS-CAPTURE INTELLIGENCE

Cryptographic fingerprints, configuration drift and correlation across captures of the same service.

TECHNICAL APPROACH

Each stage consumes the previous stage's evidence and preserves packet-level provenance.



FEASIBILITY AND VIABILITY

Feasible because it is built and measured. Every figure below was re-run on the release commit.

1,357

backend tests pass

1,367

with the TShark cross-check

89

frontend tests

5

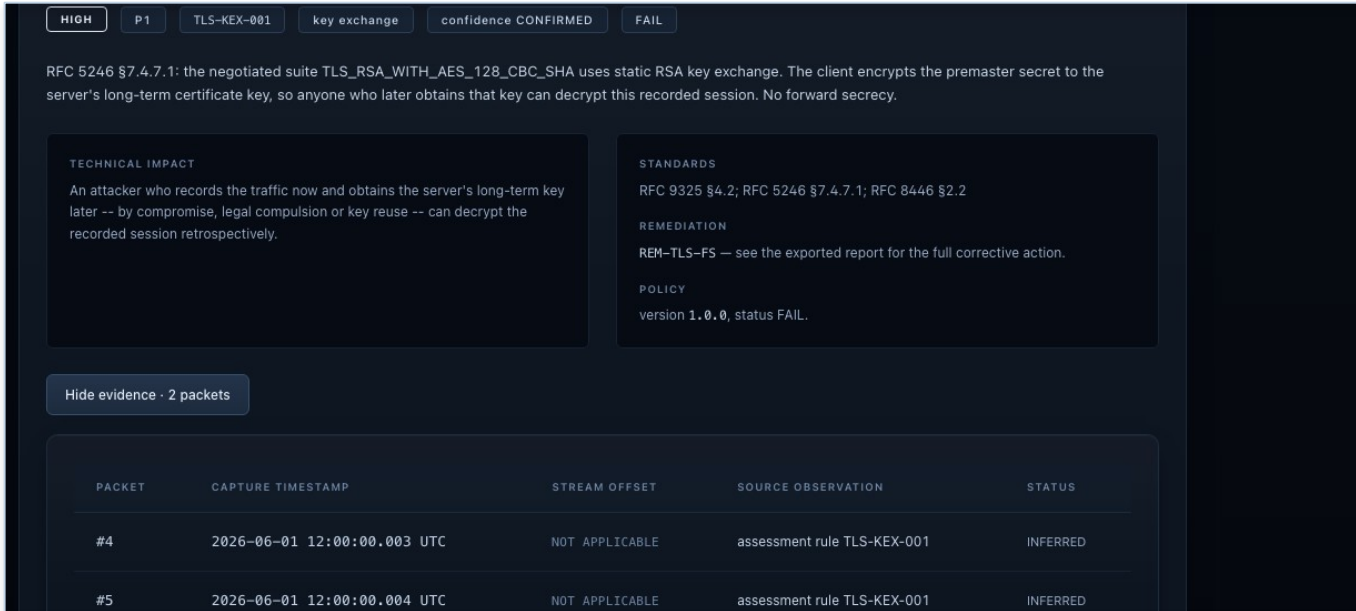
real-backend E2E specs

158

files ruff + mypy clean

0

known advisories at audit



HIGH **P1** **TLS-KEX-001** key exchange confidence CONFIRMED FAIL

RFC 5246 §7.4.7.1: the negotiated suite TLS_RSA_WITH_AES_128_CBC_SHA uses static RSA key exchange. The client encrypts the premaster secret to the server's long-term certificate key, so anyone who later obtains that key can decrypt this recorded session. No forward secrecy.

TECHNICAL IMPACT
An attacker who records the traffic now and obtains the server's long-term key later -- by compromise, legal compulsion or key reuse -- can decrypt the recorded session retrospectively.

STANDARDS
RFC 9325 §4.2; RFC 5246 §7.4.7.1; RFC 8446 §2.2

REMEDATION
REM-TLS-FS — see the exported report for the full corrective action.

POLICY
version 1.0.0, status FAIL.

Hide evidence - 2 packets

PACKET	CAPTURE TIMESTAMP	STREAM OFFSET	SOURCE OBSERVATION	STATUS
#4	2026-06-01 12:00:00.003 UTC	NOT APPLICABLE	assessment rule TLS-KEX-001	INFERRED
#5	2026-06-01 12:00:00.004 UTC	NOT APPLICABLE	assessment rule TLS-KEX-001	INFERRED

TLS-KEX-001, HIGH — and the packets it was evaluated against. Actual product output.

OBSERVE

Static RSA key exchange negotiated

**TRACE**

Packets #4 and #5, with timestamps

**ASSESS**

TLS-KEX-001 · HIGH · no forward secrecy

**REMEDiate**

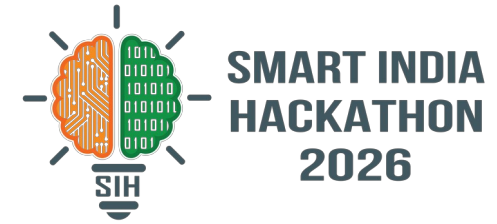
Move to ephemeral key exchange

KNOWN BOUNDARIES

TLS 1.3 may encrypt certificate evidence a passive capture cannot recover; it is reported NOT AVAILABLE with the reason, never guessed.
The supervised classifier is experimental and never overrides a deterministic finding.

Zero-Day

IMPACT AND BENEFITS



For authorised email-security investigation — and every claim below is what the product already outputs.

DISCOVER

Observable weak cryptography

EXPLAIN

Packet-backed evidence

PRIORITIZE

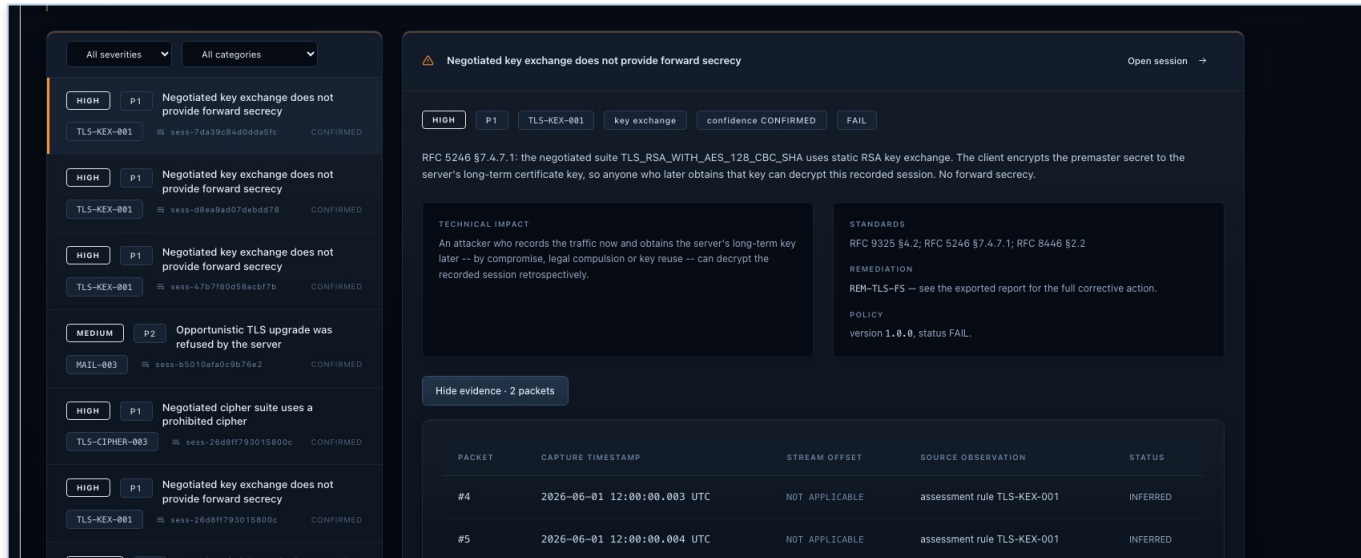
What to fix first

TRACK

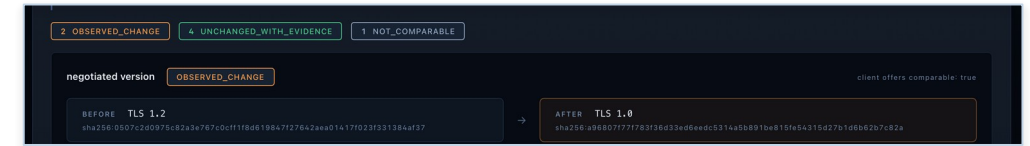
Cryptographic drift

REPORT

Portable forensic output

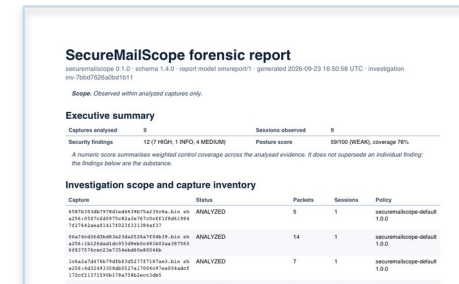


DISCOVER · EXPLAIN · PRIORITIZE — one finding, its rule, its severity and its remediation.



TRACK — the same observed service, two captures.

EARLIER TLS 1.2 → OBSERVED_CHANGE → LATER TLS 1.0



The report carries
findings · packet references ·
remediation
JSON · standalone HTML · PDF

Demonstrated on controlled synthetic captures; no production deployment claim.

REPORT — the exported PDF.

RESEARCH AND REFERENCES

The standards the rules cite, and the pipeline every claim in this deck is checked by.

CORE TLS / PKI

RFC 8446 — TLS 1.3
RFC 9325 — secure use of TLS
RFC 5280 — X.509 and CRL profile
NIST SP 800-52 Rev. 2

EMAIL SECURITY

RFC 5321 — SMTP
RFC 3207 — STARTTLS
RFC 9051 — IMAP 4rev2
RFC 1939 / RFC 2595 — POP3 and TLS

VALIDATION / TOOLING

Wireshark / TShark
Scapy
cryptography
pytest · Playwright

HOW EVERY CLAIM IS CHECKED

CONTROLLED FIXTURES

one fixed seed, identical bytes everywhere

HAND-DERIVED EXPECTATIONS

written by hand, committed, not read back

ENGINE UNDER TEST

1,357 tests; captures stay out of git

INDEPENDENT TSHARK CROSS-CHECK

against Wireshark's own dissector

REAL-BACKEND PLAYWRIGHT

five browser specs, including a restart

REPORT PARITY

JSON, HTML and PDF asserted to agree

PROJECT

GitHub repository <https://github.com/sgetsuiith141-wq/securemailscope>

Demonstration video *Included in the submission package.*



SCAN FOR THE REPOSITORY

Public repository — source, tests and submission package.